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Interpretable machine learning model for cardiovascular disease risk prediction: a feature decomposition-based study

Liliang Yu[1†], Jiancheng Wu[1†], Xin Wu[2], Chenxi Chen[3], Yuanjun Chen[4], Li Fang[4*] and Daikun Zheng[1*]

Abstract

Background Cardiovascular disease (CVD) is a leading public health issue worldwide. The key to preventing CVD is the early prediction and identification of CVD risk factors. The aim of this study is to construct and validate CVD prediction models using machine learning (ML).

Methods This study utilized 11 features from 68,205 CVD respondents in the Kaggle dataset. Experiments were conducted using a feature decomposition-based deep learning model (FDDL) to predict CVD incidence in this dataset. The proposed model was compared with six other machine learning models. Moreover, the SHAP method was employed to interpret the model in this study.

Results The FDDL model demonstrated superior predictive capability, achieving benchmark metrics of 75.52% accuracy, 78.14% precision, 71.68% recall, an F_1 score of 0.7522, and an AUC-ROC value of 0.7643. In contrast, the LR model exhibited the weakest predictive ability among the compared methods. SHAP value-based feature importance ranking identified diastolic blood pressure, cholesterol level, systolic blood pressure, and age as the most critical predictors for cardiovascular disease risk assessment in our dataset.

Conclusion We have developed an ML model for predicting the risk of CVDs. This model shows potential to assist clinicians in identifying high-risk patients and providing a theoretical basis for personalized preventive healthcare measures.

Keywords Cardiovascular disease, Machine learning, Attention mechanism, Decomposition model

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Introduction

Cardiovascular disease (CVD) is a major global public health issue and one of the leading causes of death [1]. According to the American Heart Association, CVD caused over 19.9 million deaths globally in 2021, with projections exceeding 23.6 million by 2030, imposing a significant burden on health care systems [2]. CVD is characterized by a prolonged asymptomatic phase with multiple risk factors, complicating clinical diagnosis and population health management. Early identification of high-risk individuals is therefore critical.

Clinical prevention guidelines suggest that risk stratification can benefit asymptomatic patients [3]. Personalized risk assessment enables early CVD detection, and the European Association of Preventive Cardiology (EAPC) emphasizes the utility of prediction models in population-level prevention and health promotion [4, 5]. Currently, many risk prediction models have been recommended in different guidelines. For example, the Framingham Cardiovascular Risk Score (FGCRS) [3, 6], Systematic Coronary Risk Evaluation (SCORE) [7, 8], and the QRISK score are recommended by the American College of Cardiology/American Heart Association (ACC/AHA) and the UK National Institute for Health and Care Excellence (NICE) guidelines, respectively [9–11]. However, these models rely primarily on statistical methods (e.g., logistic regression and Cox proportional hazards) with limited risk factors, and their accuracy is often compromised by data distribution biases or multicollinearity [12].

Machine learning (ML) is a branch of artificial intelligence that uses large datasets to discover novel risk predictors and model their complex interactions, thereby increasing prediction accuracy. The initial phase leverages traditional machine learning with limited yet critical clinical variables (e.g., age, blood pressure, ECG features) for diagnostic support. Titapiccolo et al. pioneered the integration of LASSO regression's feature sparsity with random forest's ensemble advantages to develop a dynamic 6-month risk prediction framework, enhancing early intervention efficacy [13]. Notably, deep learning (DL) based on artificial neural networks is a subset of machine learning. With the rapid development of DL, its powerful feature learning and large-scale data processing abilities have attracted the attention of CVD researchers. However, DL models are often opaque black boxes that lack interpretability in decision-making processes [14, 15]. Kumar et al. further fused KNN with linear SVM metrics, achieving 96% critical-case decision accuracy with only 3.10% missed diagnoses [16]. Their follow-up employed the Hawks Optimizer (HO) to coordinate stacked deep forests, overcoming class imbalance, and introduced the binary krill herd algorithm (B-KHA) to achieve an accuracy rate of 95% [17, 18]. Pujadas et al. integrated vascular risk factors (VRFs), cardiac MRI (CMR), and radiomic features into a multimodal fusion model, achieving precise discrimination of atrial fibrillation (AUC = 0.93), heart failure (AUC = 0.89), myocardial infarction (AUC = 0.91), and stroke (AUC = 0.88) [19].

In this study, our research aims to develop a feature decomposition-based deep learning (FDDL) model for CVD risk prediction. Using medical records of 68,205 patients from the Kaggle database (11 features per patient), we performed hierarchical decomposition of risk factors (e.g., physiological indicators) and integrated an attention mechanism to prioritize key predictors, thereby optimizing disease probability estimation.

Methods

Model building

The decomposition model proposed in this article is shown in Fig. 1. The model consists of a decomposition network composed of several residual blocks, an attention module and a fully connected layer.

Decomposition network

For patients' physiological data, a decomposition network is first employed to perform feature extraction. The residual decomposition architecture isolates and disentangles raw physiological features, which captures key information in the data more efficiently. Additionally, physiological features often exhibit mixed effects due to mutual interference between different characteristics. The decomposition network can disentangle these interactions, rendering each feature independent and clearly defined. Through this feature decomposition process, causal relationships and latent structures within the data become more interpretable, thereby enhancing the model's overall explainability.

First, the patient's physiological characteristic data are fed into the decomposition network for feature extraction. Each block receives an input x_i and produces an output B_i, which can be formally expressed as $B_i = Block_i(W; x_i)$.

Within the block, multiple fully connected layers are composed, which can effectively model the relationships between input features by learning their linear dependencies. The calculation formula for the linear layer is $y' = w \bullet x' + b$, where x is the input vector, y is the output vector, w is the weight matrix, and b is the bias vector.

The blocks are connected using residual connections, meaning that the input to the next block is the difference between the input and output of the previous block. Through this residual approach, the feature data are decomposed, separating features that interfere with each other, thereby making the relationships between related features more independent and clearer.

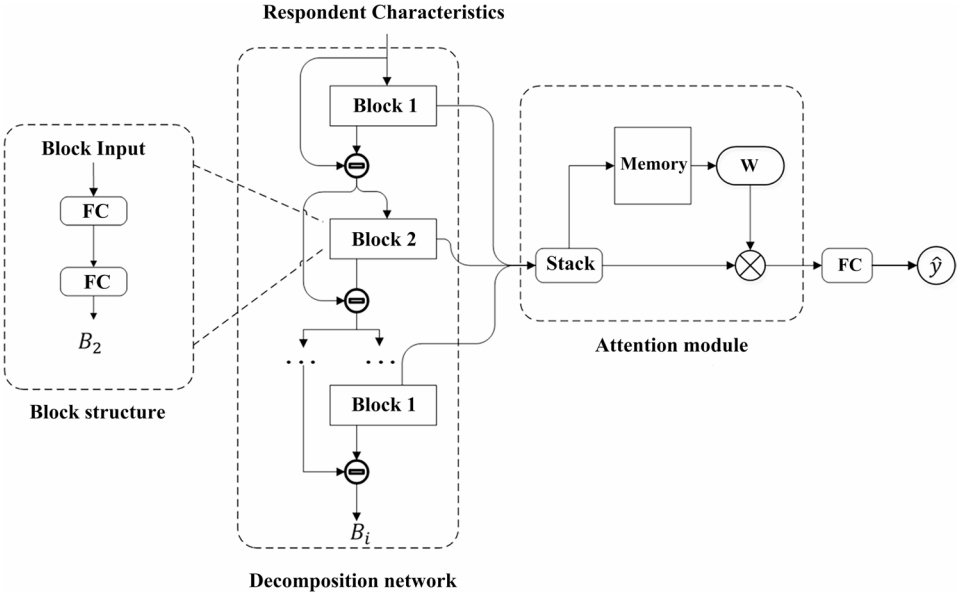


Fig. 1 Model structure diagram

The output of each block is stacked, combining feature vectors into higher-dimensional feature vectors, which are then used for further analysis of the relationships and importance of different feature combinations.

Attention module

The attention module assigns adaptive weights to the distinct feature combinations derived from decomposition, thereby enabling the model to focus more effectively on the most critical feature combinations. This mechanism facilitates the efficient identification and utilization of significant feature information, which in turn enhances the model's performance and generalizability. The data features obtained from the decomposition network are denoted as $Stack \in R^{D\times E}$, where D represents the number of decomposition layers and E denotes the dimensionality of the patient feature data. The stacked features generated by the decomposition network are then subjected to a dot product operation with the memory matrix Memory to compute the score for each layer's vector. The calculation formula is as follows: $score = dot\,(Stack, memory)$.

After the attention scores are calculated, normalization is applied to assign weights dynamically to the features at each layer. This process enables the model to prioritize important feature combinations while reducing the influence of potentially noisy or irrelevant data combinations. The calculation formula is as follows: $w\,(i) = \frac{\exp(score(i))}{\sum_i^I exp(score(i))}$

The calculated feature weights are subsequently multiplied by the stacked features, and the resulting weighted feature vectors are added together to obtain a new feature vector. The formula for this process is as follows: $feature = \sum_i^I w\,(i) * score\,(i)$

Finally, the obtained feature vector feature is sent into a fully connected layer and classified by the function $\mathrm{softmax}$. The formula is as follows: $\widehat{\mathrm{y}} = \mathrm{Softmax}(\mathrm{w_{fc}} \bullet \mathrm{feature} + \mathrm{b_{fc}})$.

Model training

We model the prediction of CVD mortality as a classification problem and use the binary cross-entropy loss function. Let y represent the data label and $\widehat{\mathrm{y}}$ represent the value predicted from the model.

We model the prediction of cardiovascular disease mortality as a classification problem and thus employ the binary cross-entropy loss function. This function is computationally efficient and straightforward when calculating gradients, and it is directly related to probability, making it highly suitable for evaluating the quality of model outputs. The formula is shown below, where y represents the data labels and where $\widehat{\mathrm{y}}$ denotes the values predicted by the model.

$$\mathrm{L}\,(\mathrm{y}, \widehat{\mathrm{y}}) = -(\mathrm{y}\log\,(\widehat{\mathrm{y}}) + (1-\mathrm{y})\log\,(1-\widehat{\mathrm{y}}))$$

To optimize the model, we use the gradient descent algorithm. It calculates the gradient of the loss function with respect to the parameters and updates the parameters in the direction opposite to the gradient. This method can find the global optimal solution in convex functions, where α is the learning rate and θ is the model parameter. $\theta' = \theta - \alpha \nabla_\theta L\,(y, \widehat{y})$

Data Source

The dataset used in the experiment is an open-source CVD dataset from Kaggle, comprising medical records of 68,205 patients. Each patient's record includes 11 features, categorized into three groups: (1) factual information (e.g., age, weight, height), (2) examination measurements (e.g., systolic blood pressure, diastolic blood pressure, blood glucose levels), and (3) subjective indicators (e.g., smoking status, alcohol consumption, and exercise habits). The dataset can be accessed at https://www.kaggle.com/code/sulianova/eda-cardiovascular-data.

Data preprocessing

Before training the model, the data underwent a thorough preprocessing process. First, the numerical features were standardized using StandardScaler, which transforms all the input features into a standard normal distribution with a mean of 0 and a standard deviation of 1, thereby eliminating the impact of varying feature scales on model performance. Categorical variables were encoded as numeric values using LabelEncoder to ensure that they could be used as inputs to the neural network. Additionally, missing values in the dataset were identified and handled using mean imputation for numerical features, ensuring the integrity of the data. Finally, the data were split into training and testing sets using train_test_split, with an 80–20 split. Stratified sampling (stratify parameter) was applied to maintain the class distribution (i.e., the presence or absence of cardiovascular disease) across both the training and testing sets, enhancing the model's ability to generalize.

To solve the problem of imbalanced data categories, we adopted the following methods: (1) Weighted loss function: A class-weighted cross-entropy loss was used to penalize misclassification of the minority class more heavily. The loss weights were empirically set to [1.0, 1.1], emphasizing positive (cardio = 1) cases. (2) Synthetic oversampling: We employed the synthetic minority oversampling technique (SMOTE) to artificially increase the presence of minority class samples in the training set. SMOTE was applied only after the train–test split to avoid data leakage. To improve model generalizability further, Gaussian noise N(0, 0.01) was added to numeric features such as systolic/diastolic blood pressure, age, and cholesterol. Additionally, feature permutation within the same class was performed on a small percentage (5%) of training samples to simulate plausible physiological variability. These steps were implemented during the training pipeline and combined with normalization, encoding, and stratified sampling.

Model explanation

The interpretation of the prediction model is performed by SHapley Additive exPlanation (SHAP), which precisely calculates the contribution and influence of each feature on the final prediction. In addition, each observation in the dataset can be explained by a specific set of SHAP values.

Statistical analysis

Six machine learning models, including decision tree (DT), logistic regression (LR), support vector machine (SVM), random forest (RF), backpropagation neural network (BPNN), and light gradient boosting machine (LightGBM), were compared with our proposed predictive model. To ensure the repeatability and clarity of the proposed FDDL model, its key architecture and training parameters are listed in Supplementary Table 1. The performance evaluation metrics included the classification accuracy, precision, recall, F1 score, and area under the ROC curve (AUC). The dataset was partitioned into an 80–20 split for the training and validation sets, with model training conducted on the former and performance validation on the latter.

The data were analyzed using IBM SPSS Statistics v25.0. The Kolmogorov–Smirnov test was used for continuous variables. Continuous variables with a normal distribution are presented as the means ± standard deviations and were compared using t tests. Continuous variables with skewed distributions are presented as medians with interquartile ranges and were compared with the Mann–Whitney U test. Categorical variables are presented as numbers with percentages and were compared with the chi-square test. $P < 0.05$ was considered statistically significant.

Results

Baseline characteristics

Table 1 shows the basic characteristics of the 68,205 respondents in the database included in this analysis. In our study, patients with CVD were significantly older and had greater body weights, systolic blood pressure, diastolic blood pressure, cholesterol, and blood glucose levels than non-CVD patients were ($P < 0.05$); concurrently, individuals engaging in regular physical activity presented a lower incidence of CVD.

Model construction and evaluation

For the training dataset, we constructed the FDDL model and compared it with the DT, LR, SVM, RF, BP, and LightGBM models. As summarized in Table 2, the FDDL model demonstrated superior predictive capability, achieving benchmark metrics of 75.52% accuracy, 78.14% precision, 71.68% recall, an F1 score of 0.7522, and an AUC-ROC value of 0.7643. In contrast, LR exhibited

Table 1 Comparison of baseline characteristics between the non-CVD and CVD groups

Variable	Cardiovascular diseases			*P*
	Total (*n* = 68205)	CVDs	Non-CVDs	
Age	53(48,58)	55(50,60)	52.00(46,57)	< 0.001
Sex				0.11
Female	44,427(65.10%)	21,834(64.84%)	22,593(65.42%)	
Male	23,778(34.90%)	11,838(35.16%)	11,940(34.58%)	
Height	164.37 ± 8.18	164.28 ± 8.23	164.46 ± 8.12	< 0.001
Weight	74.10 ± 14.29	76.67 ± 14.83	71.59 ± 13.27	< 0.001
Systolic pressure	126.43 ± 15.96	133.45 ± 16.40	119.60 ± 12.10	< 0.001
Diastolic pressure	81.26 ± 9.14	84.44 ± 9.19	78.17 ± 7.96	< 0.001
Cholesterol				< 0.001
Normal	51,222(75.10%)	22,269(66.14%)	28,953(83.84%)	
Above normal	9191(13.48%)	5470(16.24%)	3721(10.78%)	
Well above normal	7792(11.42%)	5933(17.62%)	1859(5.38%)	
Blood glucose				< 0.001
Normal	58,027(85.08%)	27,550(81.82%)	30,477(88.25%)	
Above normal	4998(7.33%)	2930(8.70%)	2068(5.99%)	
Well above normal	5180(7.59%)	3192(9.48%)	1988(5.76%)	
Smoking				< 0.001
Yes	5979(8.77%)	2792(8.29%)	3187(9.23%)	
No	62,226(91.23%)	30,880(91.71%)	31,346(90.77%)	
Drinking				< 0.001
Yes	3624(5.31%)	1720(5.11%)	1904(5.51%)	
No	64,581(94.69%)	31,952(94.89%)	32,629(94.49%)	
Active				< 0.001
Yes	54,806(80.35%)	26,543(78.83%)	28,263(81.84%)	
No	13,399(19.65%)	7129(21.17%)	6270(18.16%)	

Table 2 Performance of each model for prediction

Model	Accuracy (%)	Precision (%)	Recall (%)	F_1 score	AUC
DT	72.97	75.51	68.13	0.7163	0.7298
LR	72.35	74.56	68.00	0.7113	0.7236
SVM	73.19	74.72	70.25	0.7241	0.7320
RF	73.69	76.70	68.17	0.7218	0.7370
BPNN	73.48	77.64	67.87	0.7292	0.7364
Light GBM	73.93	76.30	69.55	0.7277	0.7394
FDDL	75.52	78.14	71.68	0.7522	0.7643

the lowest performance among the compared models. The model convergence analysis revealed a progressive improvement in predictive efficacy with increasing training epochs. As illustrated in (Fig. 2A), the classification accuracy stabilized at approximately 75% by the 40th epoch, ultimately reaching peak performance (75.52%) at the 100th training iteration. Additionally, the model achieved a high true-positive rate (TPR) and a low false-positive rate (FPR), further confirming its superior predictive accuracy (Fig. 2B).

Explanation of the FDDL model with the SHAP method

The SHAP method was employed to evaluate the importance of each predictive variable in the FDDL model's predictions. This study modeled the relationships and relative importance between features using a decomposition structure, and the feature importance analysis chart ranked the most significant variables in descending order. As shown in (Fig. 3), the most important feature was age, followed by systolic blood pressure, weight, and cholesterol. Additionally, to investigate the positive and negative relationships between predictors and the target outcome, SHAP values were utilized to identify disease risk factors. As illustrated in (Fig. 4), negative values on the horizontal axis indicate negative correlations between the input features and the outcome, whereas positive values indicate positive correlations; the magnitude of the SHAP value reflects the strength of the correlation. The size of the feature points is represented by a color gradient ranging from blue (small) to red (large). The results demonstrated that increases in diastolic blood pressure, cholesterol, systolic blood pressure, and age were significantly positively correlated with the occurrence of the disease, thereby promoting its progression.

Explanatory analysis of a single sample via SHAP

The SHAP method was employed to visualize the interpretability of each input feature from the base value to the model output, where red indicates that the feature increases the predicted probability and blue denotes a decrease in the predicted probability. As illustrated in (Fig. 5), compared with other indicators, the systolic blood pressure in the sample data significantly increased the prediction probability.

Discussion

Artificial intelligence and machine learning are driving medical advancements, offering novel perspectives for the prediction and diagnosis of CVDs [20–22]. In this study, we developed an FDDL (feature decomposition deep learning) model and applied it to the prediction of CVDs. Compared with traditional machine learning models, the proposed FDDL model exhibited superior predictive performance (accuracy: 75.52%, AUC = 0.7643), indicating that the FDDL model can capture the key information in patients' health records more accurately.

Although FDDL models demonstrate significant advantages in terms of prediction accuracy, traditional machine learning models (e.g., DT, RF) maintain

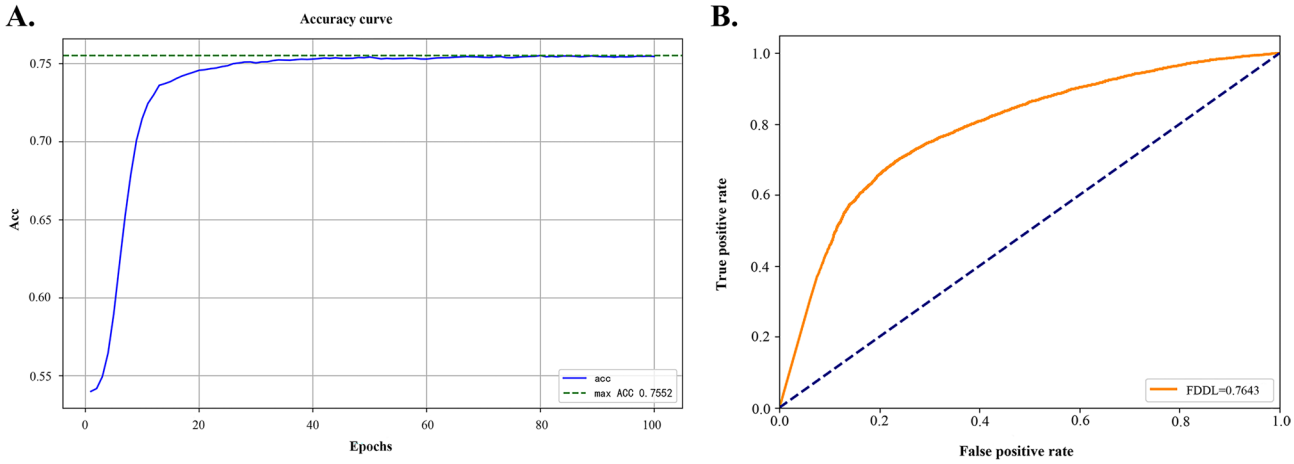


Fig. 2 Accuracy curve and ROC curve of the FDDL model. (**A**) Accuracy curve. (**B**) Receiver operating characteristic curve

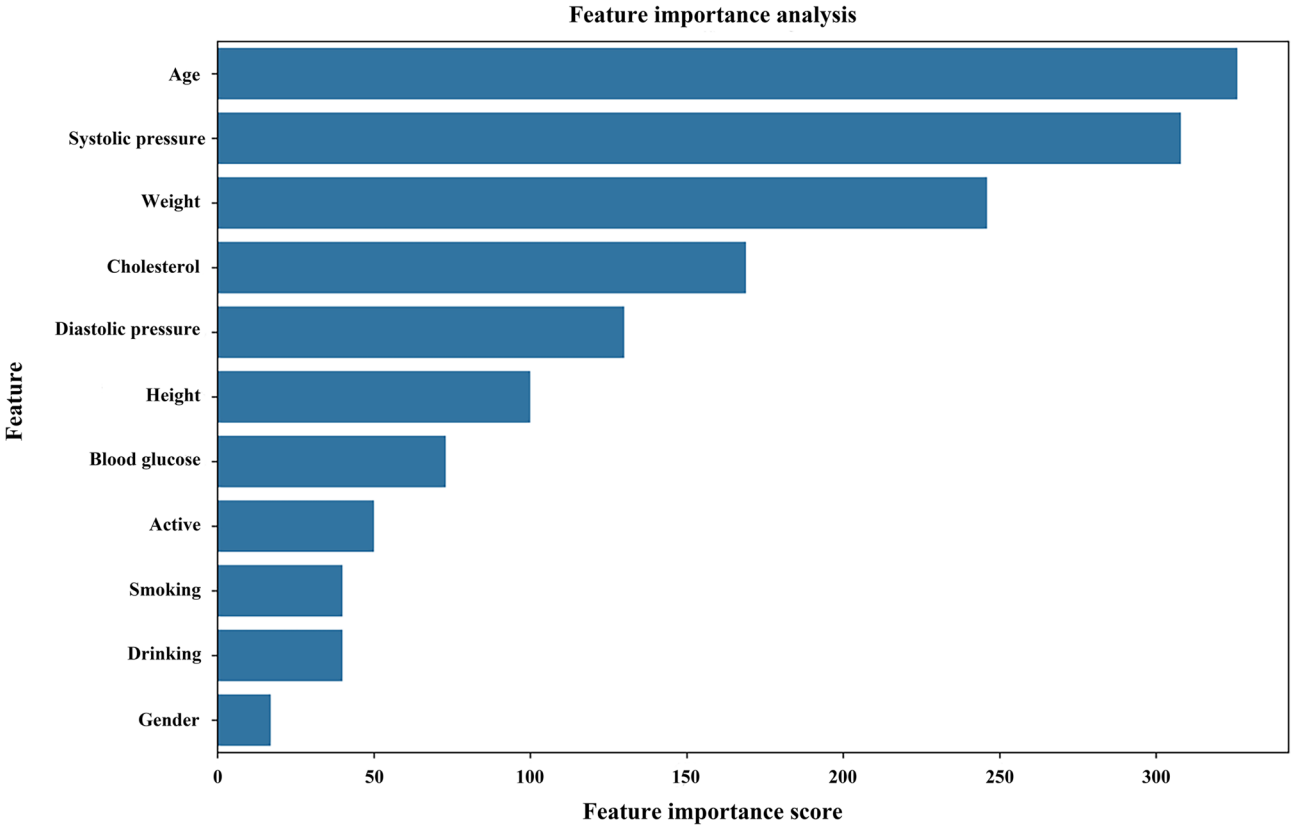


Fig. 3 Characteristics of the variable importance analysis

superior interpretability [23, 24]. To address this limitation and explore the interpretability of FDDL models, we employed SHAP analysis to elucidate the model's decision-making logic. The SHAP results identified age, systolic blood pressure, and cholesterol as key predictors of CVDs, which is consistent with the FGCRS, thereby validating both the rationality and reliability of the model's predictions [3, 6]. This alignment reinforces the clinical validity of our model and its compatibility with existing clinical reasoning pathways. Additionally, in the dataset, the overall class distribution of the target variable (cardio) was nearly balanced (0: 50.03%, 1: 49.97%), and several predictor variables exhibited severe internal class imbalance. For example, 90.4% of the samples had normal cholesterol levels, and over 90% were nonsmokers. These skewed distributions of key risk factors can lead the model to underrepresent rare subgroups during training. Therefore, we adopted a method including

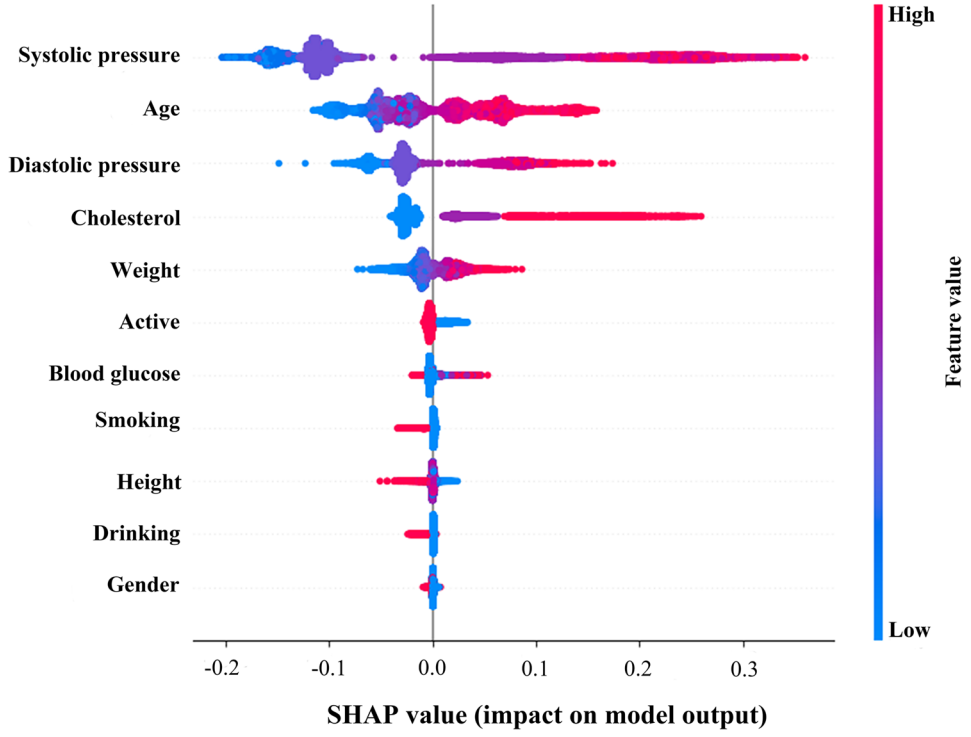


Fig. 4 Interpretability analysis of the model by the SHAP method

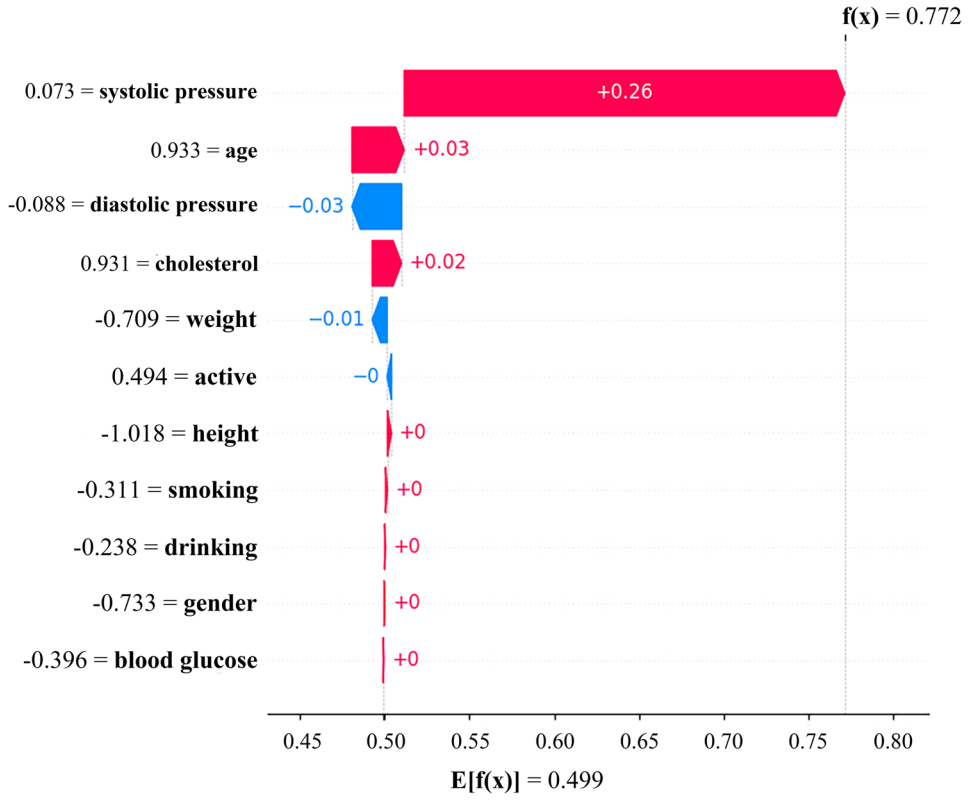


Fig. 5 SHAP single-sample plot

a weighted loss function and synthetic oversampling to address these challenges.

During model training, we observed that network decomposition effectively reduced feature aliasing effects, thereby clarifying causal relationships between individual features, which supports the potential of this model in clinical applications. However, this study has the following limitations. First, our data were derived from the Kaggle dataset; thus, it was very difficult to completely avoid bias in the data, resulting in insufficient representativeness of the data. Second, owing to the inherent limitations of the database, we evaluated 11 available clinical feature predictors. These variables are all routine items that are easily accessible in clinical practice, which may have led to the omission of some key variables in our analysis. To improve the accuracy of these predictive models, other parameters, such as details of drug use, dosage, quantification of smoking frequency, and other lifestyle indicators, need to be considered. Molecular markers related to CVDs and other medical indicators also need to be included. Third, due to the limitations of space and computing resources in this study, the complete ablation experiment was not included in the current version. However, we still believe that the good performance of the model has not been masked. But we also recognize that external verification is needed to ensure universality. In future research, we will conduct a complete ablation study. Fourth, although the dataset did not permit explicit subgroup analysis (e.g., by gender or ethnicity), the SHAP dependence plots reveal varying risk associations across feature values, suggesting that the model captures heterogeneous risk patterns. In future multicenter studies, we will further clarify subgroup-specific risk patterns. Fifth, the data came from a single center. Further large-scale multicenter studies are needed to verify and adjust the accuracy and practicability of our model, and external validation should be conducted on datasets from different institutions or populations. In the future, we will conduct multicenter research to obtain larger-scale and more representative data and improve the generalizability of the model. Meanwhile, herein, various feature extraction techniques were evaluated to determine the best method for data representation.

Conclusions

In conclusion, we have developed an ML model for predicting the risk of CVDs. This model shows potential to assist clinicians in identifying high-risk patients and providing a theoretical basis for personalized preventive healthcare measures.

Abbreviations

CVD	Cardiovascular diseases
AUC	Area under the curve
FDDL	Feature decomposition-based deep learning
EAPC	European Association of Preventive Cardiology
FGCRS	Framingham Cardiovascular Risk Score
SCORE	Systematic Coronary Risk Evaluation
ACC	American College of Cardiology
AHA	American Heart Association
NICE	National Institute for Health and Care Excellence
SHAP	SHapley Additive exPlanation
DT	Decision Tree
LR	Logistic Regression
SVM	Support Vector Machine
RF	Random Forest
BPNN	Backpropagation Neural Network
Light GBM	Light Gradient Boosting Machine

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-025-24921-4.

Supplementary Material 1.

Supplementary Material 2.

Acknowledgements

Not applicable.

Authors' contributions

LLY drafted the manuscript. LLY, JCW, Chenxi Chen collected and analyzed the data. JCW, XW, built the model. LLY, YJC, LF, DKZ designed the study. All authors contributed to the article and approved the submitted version.

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Data availability

The dataset used and/or analyzed during the current study is available from the corresponding author upon reasonable request or accessible at the following URL: https://www.kaggle.com/code/sulianova/eda-cardiovascular-data.

Declarations

Ethics approval and consent to participate

Ethical approval and individual patient consent was not necessary because all the protected health information was anonymized.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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